Full-Stack Web Portfolio

Project Report — Web Programming Course

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1. Project Overview

This project is a dynamic, full-stack personal portfolio website designed and developed as part of the Web Programming course. The goal was to create a professional online presence that demonstrates proficiency across the complete web development stack — from semantic HTML and responsive CSS to server-side PHP, MySQL databases, and client-side JavaScript. The portfolio showcases real projects, provides a working contact form, and includes a secure admin dashboard for content management.

2. Technology Stack

Layer	Technology	Usage
Frontend	HTML5, CSS3	Semantic structure, responsive layout
Styling	CSS Flexbox & Grid	Mobile-first responsive design, dark/light mode
Fonts	Google Fonts (Syne + DM Sans)	Display + body typography via CDN
Client Logic	JavaScript (ES6+)	DOM manipulation, validation, AJAX, cookies
Backend	PHP 8+	Server-side logic, session management, AJAX endpoints
Database	MySQL (mysqli)	Projects & contacts storage, prepared statements
Server	Apache (XAMPP)	Local development environment

3. Key Features

RESPONSIVE DESIGN

Mobile-first layout using CSS Grid and Flexbox with three breakpoints (480px, 720px, 900px). Adapts seamlessly from mobile to desktop.

DARK / LIGHT MODE

Theme toggle persisted via a 365-day cookie. Applied via CSS custom properties for instant, flicker-free switching without page reload.

DYNAMIC PROJECTS (AJAX)

Projects are stored in MySQL and fetched asynchronously via the Fetch API from get_projects.php, which returns JSON. Cards are injected into the DOM without any page refresh.

CONTACT FORM

Client-side JavaScript validation (required fields, email regex, minimum length) combined with server-side PHP validation and sanitization. Messages are saved to the contacts table in MySQL using prepared statements.

ADMIN DASHBOARD

Session-based login system with session_regenerate_id() on authentication. Full CRUD operations on projects (add, edit, delete) and a read-only messages table. Logout destroys the session and clears the remember cookie.

SCROLL ANIMATIONS

IntersectionObserver API triggers CSS reveal transitions as elements enter the viewport, providing a smooth, performant scroll experience.

SECURITY

All user inputs are sanitized with htmlspecialchars() on the PHP side and escHtml() on the JavaScript side. Prepared statements with bind_param() prevent SQL injection.

4. Project Structure

File / Folder	Description
`index.php`	Main entry point — hero, about, projects, skills, contact sections
`css/style.css`	External stylesheet with CSS variables, dark mode, responsive breakpoints
`js/main.js`	All client-side logic: dark mode, AJAX, form validation, animations
`php/db.php`	Singleton database connection (mysqli)
`php/get_projects.php`	AJAX endpoint — queries projects table and returns JSON
`php/contact.php`	AJAX endpoint — validates and saves contact form data to MySQL
`admin/login.php`	Admin authentication with session management
`admin/dashboard.php`	CRUD panel for projects; read-only view for contact messages
`admin/logout.php`	Session destroy and cookie cleanup
`sql/portfolio_db.sql`	Full MySQL export — tables (projects, contacts) and sample data

5. Requirements Coverage

Requirement	Implementation	Status
Semantic HTML5 tags	nav, section, footer, form throughout index.php	**Done**
Tables & Forms	Contact form + admin dashboard tables	**Done**
CSS Box Model / Flexbox / Grid	Used throughout layout and responsive breakpoints	**Done**
External stylesheet	css/style.css with CSS custom properties	**Done**

Requirement	Implementation	Status
Dark mode / dynamic UI	Cookie-persisted theme toggle in main.js	**Done**
JS Form Validation	validateForm() — all fields, email regex, min length	**Done**
DOM Manipulation	Scroll reveal, mobile nav, project card injection	**Done**
Contact saved to MySQL	php/contact.php + contacts table, prepared statements	**Done**
Dynamic content from DB	php/get_projects.php returns JSON, injected via Fetch API	**Done**
AJAX / Fetch API	loadProjects() + contact form submit — no page reload	**Done**
Sessions & Cookies	Admin session login; theme + remember cookies	**Done**
Admin Dashboard (CRUD)	admin/dashboard.php — add, edit, delete projects	**Done**
SQL export file	sql/portfolio_db.sql included in ZIP	**Done**

6. Challenges & Solutions

One of the main challenges was ensuring the dark mode toggle worked without a flash of unstyled content on page load. This was solved by reading the theme cookie synchronously in JavaScript before the DOM painted, applying the class immediately. Another challenge was keeping the AJAX project loading smooth while also re-triggering the IntersectionObserver reveal animations for dynamically created cards — resolved by calling observer.observe() on each card with a staggered setTimeout. The admin dashboard required careful session security: session_regenerate_id(true) is called on every successful login to prevent session fixation attacks.

7. Conclusion

The project successfully fulfills all technical requirements of the Web Programming course. It demonstrates a complete, integrated full-stack workflow: a responsive and accessible frontend, asynchronous data fetching, server-side validation, relational database storage, and a secure admin interface. Beyond the coursework requirements, the portfolio serves as a real professional asset ready to be shared with potential employers.